The present and future of blockchain and crypto

Maddison Turner

23064847

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1 Introduction

Blockchain and cryptocurrency has grown a large interest in the last decade due to how it is managed and how it can be implemented. This report explains what blockchains and cryptocurrencies are and how they are used in the real world, as well as delving deeper into the ethical background of them and how they are constructed. It also covers how they are used in different sectors, for example: Finance and real estate. In addition, how they may be used in the future.

2 Technical overview

2.1 Blockchain Technologies

Distributed ledger technology, also known as DLT, is an infrastructure that allows simultaneous access, validation, and updating of records across a network. With the use of DLT it allows users within the network, who may not trust each other, to interact without the need for a third party authoriser. This adds a level of security, traceability, and transparency within the network. DLT can be programmed to be immutable, meaning once a transaction is recorded the data that is stored cannot be changed. DLs work by each member on the network having a copy of the database. If a member updates the database then that is communicated across the network (El Ioini & Pahl 2018).

Pros of DLT

- Decentralized- since the ledgers aren't controlled by one person nobody can change the data or recorded transactions.
- Security- distributed ledgers use cryptographic principles to keep data safe, the network verifies each transaction which makes it extremely hard to tamper the data.
- Immutability- once a transaction is recorded it cannot be changed or deleted.
- Flexibility- distributed ledgers can be programmed to fit the needs of the industry.
- Speed- since distributed ledgers do not need third party authorisation this means transactions can be completed much quicker, perfect for financial or IT sectors.

Cons of DLT

- Scalability- the bigger the network grows the harder it is to establish a consensus making the time for each transaction increase, therefore slowing down the network.
- Energy consumption- since the network is made up of a large number of nodes to validate transactions this means a lot of power is needed. This is extremely bad for the environment.
- Regulation- since there is no third party authoriser then it makes it hard to regulate. This makes it hard to prove the technology is safe to use.
- Accessibility- distributed ledgers can be hard to understand which can make it difficult to implement.

(Aryan 2024)
The opposite of distributed ledger technology is a centralized ledger. This is a type of database that is controlled by a single authority, like a company or government. All the data is stored

in one location that can only be accessed by the authoriser or someone that has been granted access from them. This type of ledger is commonly used in the financial sector. It allows security for customer's transactions histories, account balances, and finanacial information like credit scores. This type of ledger has a good level of security as if any information got out then the leak can be tracked back to who ever has access to it. However, this system does have one big downside; any hacker that gains access to the centralized ledger can leak all of the data within it.

Pros of CLT

- Data control- centralized ledgers provide greater data control for organisations since they can meet data handling regulations easier.
- Simple data control- since the data is all kept in one place and managed by one individual it makes retrieval of the data quick, easy, and reliable.
- Cheap- because the data is managed by a single individual it makes: costs lower as it takes less to manage, maintain, power, and there is no need for extra people or infrastructure.

Cons of CLT

- Single point failure- if the ledger becomes compromised this means that, anyone that relies on the data it withholds may be disrupted.
- Accessibility- due to the data being managed by a single individual, it can cause others that need access to the data to be denied.
- Prone to attacks- while these ledgers are strong they are extremely prone to cyber attacks making them vulnerable. Any hacker that gains access to the data may be able to leak it causing it to be compromised. This can sometimes lead to large lawsuits and fines upon companies that use centralized ledgers.

(Aryan 2024)
Blockchain is a distributed ledger which is shared and maintained across computer network nodes. Blockchain is known for its crucial role in cryptocurrency systems, due to its secure and decentralized record of transactions. Also, since blockchain is a distributed ledger this means that data immutable (Hayes 2025). (Aaron.S 2025) describes a blockchain platform as a digital infrastructure that can be used to streamline various processes for blockchain-based applications and services, because of this there are several different blockchain platforms that are in use today.

2.2 Properties of blockchains

A blockchain is made up of a number of blocks, digital containers used to permanently record data to do with transactions on a network. Each blockchain begins with first block which is called the genesis block, also known as block 0. A basic genesis block must contain the relevant properties: index, timestamp, hash, and some data. The genesis block is very important to the blockchain as it initializes the network, sets the block price, and ensures a consensus. The blocks that come after the genesis block must include the previous hash as well as the properties within the genesis block.

For some blockchains there needs to be an additional property known as proof-of-work (PoW). This is a consensus mechanism which chooses who can add the next block to the blockchain.

For example, when mining for bitcoin a PoW requires miners to solve a complex mathematical puzzle before they can add a new block to the chain. This is necessary as it validates that the miner has put in the time and energy before they get their reward. This also prevents double spending and making sure each bitcoin is used only once. This system is very reliable as it is highly unlikely that someone could tamper with the blockchain. This is because if someone did want to change the information in a block they would have to also change the information in the previous block, but to do this all of the computers participating in the network would have to agree to the changes. To do all of this the computational power, time and costs would more than likely out-way the reward (Montevirgen 2025).

2.3 What is are Cryptocurrencies

Cryptocurrencies are virtual currencies that have volatile worth. In basics this means that the price of the token could be worth almost nothing way day then the next it could be worth thousands. They rely on cryptography for security and work through the use of decentralized networks like blockchain. Due to this they are not regulated by any government, however government agencies like SEC routinely set laws and regulations for crypto exchanges and online brokerage firms to follow (Cierra Murry 2025).

#	Name	Price	1h %	24h %	7d %	Market Cap
1	Bitcoin BTC	$90,714.85	▲0.54%	▲0.41%	▼1.08%	$1,811,998,298,096
2	Ethereum ETH	$3,108.22	▲0.53%	▲0.88%	▼1.21%	$375,146,152,372
3	Tether USDT	$0.9987	▲0.02%	▲0.02%	▼0.08%	$186,724,216,572
4	XRP XRP	$2.06	▲0.60%	▼0.97%	▼1.23%	$125,446,799,389
5	BNB BNB	$903.53	▲0.39%	▼0.32%	▲0.75%	$124,447,608,258
6	Solana SOL	$139.10	▲0.94%	▲2.54%	▲3.56%	$78,534,460,902
7	USDC USDC	$0.9997	▲0.02%	▲0.03%	▲0.00%	$74,628,869,979
8	TRON TRX	$0.2990	▲0.00%	▼1.00%	▲1.78%	$28,320,144,329
9	Dogecoin DOGE	$0.1375	▲0.76%	▼1.29%	▼8.16%	$23,148,639,842
10	Cardano ADA	$0.3898	▲0.86%	▲0.56%	▼2.83%	$14,018,647,784

Figure 1: Top 10 Cryptocurrencies

Above are the top ten cryptocurrencies that are currently circulating the market. One of which is Bitcoin. This cryptocurrency in particular has drawn a lot of attention. This is because it was one of the first token available to mine and trade. At the start of it's popularity it wasn't worth a lot and people didn't really understand how it worked. For example, there is a story of someone paying another person to buy them two pizzas for ten thousand bitcoin in 2010. The price of bitcoin was nowhere near the total price for the two pizzas, however if that same transaction was made today the pizzas would be worth 672,143,610.7 GBP.
Popularity for cryptocurrencies have gained more over the last five years due to online platforms like TikTok and X (formerly known as Twitter). This can be seen in the cryptocurrency known as Dodgecoin. This token went viral for its Shiba Inu mascot on the coin and the meme that it represents; along with added help from celebrities like Elon Musk this token increased in price rapidly.

3 Ethics

3.1 Environmental impact

There are multiple ethical dilemmas that need to be taken into consideration when it comes to using blockchain technologies. One major issue that needs to be tackled is the environmental impacts. Blockchain technologies are computer based meaning a large amount of wattage is needed to not only power the servers that are being used but also to keep them cool. In the case of Bitcoin, for example, in order to reach the validity of a proof of work, a billion watts is estimated to be necessary (Kirchschlaeger 2024). This is extremely unethical as a large amount of power is being wasted when it could be used elsewhere, like heating the homes of the impoverished.

3.2 Crime

morally wrong- crypto scams, money laundering Cryptocurrencies are used in many online scams and money laundering operations. This is due to its pseudonymity it is perfect for criminals to stay anonymous as they don't need to share personal information like their names but instead their wallet address. It also allows money to be sent quickly anywhere in the world without the need for banks. For these reasons cryptocurrencies are being seen more often in organised crime like money laundering. Not only that but since crypto is easily available to most people, scammers are being seen using it far more often. This because once the funds are sent the victims are unable to recover them.

4 Application Analysis

4.1 Finance

Blockchain technology is commonly used within the finance sector because of its ability to provide smart contracts, simple payment processing, loyalty programs, advanced trading/investing, and upgraded digital ID management (Sam Daley 2025). One of the most useful application of blockchain in finance are smart contracts. These are like physical contracts but the contract is fulfilled in real-time via the blockchain. This is extremely useful for the finance sector as there is no need for a third party to authorise the contract meaning additional security and increased speed.

4.2 Real estate

Blockchain is used in real estate as a way of keeping records safe and secure. However, blockchain is growing in this industry for the same reason in the finance industry, smart contracts. It represents the future of real estate where buying and selling houses or land can be as easy as online shopping (Schlapkohl 2025). In addition, when buying real estate trust is a large part of the process. The use of blockchain allows trust between the buyer and the seller as well as safety and security due to the lack of a third party to authorise the sale.

5 Code summary

I have attached my blockchain code below in the appendices, however I will use this space to give a brief explanation of my code and how it works:
I was instructed to create a blockchain code that has a length of six blocks, including the genesis

block. For each block I have printed the: index, timestamp, data, new SHA-256 hash, and the previous SHA-256 hash. The libraries that I have used are 'time' and 'hashlib', this allowed me to produce the time stamp and the SHA-256 hashes.

6 Conclusion

This report has explored what blockchain technology and cryptocurrencies are as well as how they work. Including their technical properties like proof of work consensus mechanisms. Blockchain has been explained as being a form of distributed ledger technology, which provides immutability, automation, decentralisation, and transparency. This report has also looked over ethical concerns like the impact of blockchain use on the environment and how it is being used in organised crime, scams and money laundering.

In the future, blockchain and cryptocurrency systems may be impacted by the advancements in quantum computing. This is because quantum computing can use brute force attacks to break cryptographic algorithms, if this happens than blockchain could become almost useless. Due to this greater research is being done to find quantum-resistant cryptography, looking into cryptographic methods that can withstand quantum attacks.

In conclusion, blockchain and cryptocurrencies have gained mass popularity and are evolving the way that different sectors work. They offer clear benefits but also introduce ethical problems that mustn't be ignored.

Appendices

https://github.com/maddi860/block-chain

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